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Review

The Outlook for Sustainable Development Goals in Business and Management: A Systematic Literature Review and Keyword Cluster Analysis

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Abstract: Sustainability is imperative for many countries and organizations, thus, in 2015 the UN proposed its 2030 Sustainable Development Goals (SDGs). These concrete goals put pressure on businesses to initiate sustainability practices, depending on the nature and characteristics of the business. By understanding the current state of sustainability and SDG studies in the business and management disciplines, this study aims to anticipate the outlook of SDGs in these fields. Therefore, this study conducted a Systematic Literature Network Analysis (SLNA) using 237 publications between 2015 and 2021. Based on a keyword analysis, an overview of the trends in SDG studies in business and management emerged in five clusters: technology and innovation, education and human resource management, CSR and firm performance, supply chains and governance, and business strategies. This implies that the business and management disciplines recognize a need to adopt and consider all aspects of sustainability to achieve the UN's 17 SDGs.

Keywords: sustainable development goals; sustainability; systematic literature network analysis; main path analysis

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1. Introduction

With the growing public interest in and demand for attention toward social concerns, many countries and organizations are taking the initiative to implement the Sustainable Development Goals (SDGs) developed by the United Nations [1,2]. As pressure on businesses to be socially and environmentally responsible intensifies, numerous organizations endeavor to support the SDGs, and multinational corporations have begun implementing initiatives to this end [3,4]. However, it has been challenging to incorporate the Sustainable Development Goals into business strategies, as many businesses concentrate on short-term goals rather than on the long-term perspective a focus on sustainability requires [5]. More importantly, some companies have reported negative attitudes toward the direct implementation of sustainability, as it requires significant contributions from stakeholders and considerations of corporate governance, non-financial regulations, and the external environment [6,7]. Notably, the SDGs do not provide an evaluative measurement, which makes it challenging to implement and holistically pursue these goals [8].

Due to the extensive complexity of sustainability, research in this area has evolved into different directions and suggestions concerning a wide range of disciplines, from sociology to engineering and geology to mathematics, each of which suggests diverse practices and applications. Several studies [9–12] have provided assessment frameworks for the social and environmental areas of the SDGs on supply chain and materials selection, building performance value, climate change initiatives, and technology and innovation tools. Though numerous studies have been conducted on sustainability and sustainable development, fundamental questions remain as to whether we are heading in the right direction to achieve the SDGs.

Sustainability and Sustainable Development Goals in Business and Management

The sustainability movement has gained significant attention in many different industries and also in academia. Still, our society is slow to transform due to restrictions on technological progress, but the ecological system has limited capabilities to support our massive carbon footprint [13]. The Brundtland Commission defined sustainable development as development designed "to meet the needs and aspirations of the present without compromising the ability to meet those of the future and improvement" [14]. The United Nations thus proposed the SDGs in 2015 to eradicate poverty, enhance healthcare and education, reduce inequality, and stimulate economic growth [1]. Under the five essential components (e.g., people, prosperity, peace, partnerships, and the planet), the SDGs outline 17 goals specified into 169 targets (Table 1). Aiming to achieve sustainable development by 2030, the SDGs provide objectives and indicators that measure and address various global civilization challenges "with each government setting its national targets guided by the global level of ambition" [2]. This indicates that the SDGs should be executed globally and that all countries and sectors become accountable for their execution, incorporating public, social, and private sector organizations [15,16]. By providing an integrated environmental and social framework, the SDGs aim to administer trade-offs while amplifying target interactions [17]. To pursue the SDGs holistically, the three pillars of economic, environmental, and social sustainability have to be embraced to a great extent by multiple stakeholders. Since sustainable development and sustainability are interchangeable, this is a collective societal process involving various stakeholders embracing social, environmental, and economic aspects [13].

Table 1. 17 Sustainable Development Goal.

Goal	Description	Goal	Description	Goal	Description
SDG 1 SO	No Poverty	SDG 7 SO	Affordable and Clean Energy	SDG 13 SO	Climate Action
SDG 2 SO	Zero Hunger	SDG 8 SO	Decent Work and Economic Growth	SDG 14 SO	Life Below Water
SDG 3 SO	Good Health and Well-being	SDG 9 SO	Industry, Innovation, and Infrastructure	SDG 15 SO	Life on Land
SDG 4 SO	Quality Education	SDG 10 SO	Reduced Inequalities	SDG 16 SO	Peace, Justice, and Strong Institutions
SDG 5 SO	Gender Equality	SDG 11 SO	Sustainable Cities and Communities	SDG 17 SO	Partnerships for the Goals
SDG 6 SO	Clean Water and Sanitation	SDG 12 SO	Responsible Consumption and Production		

Accordingly, the environmental aspect of sustainability focuses on quality-of-life issues such as clean air and overall health. Still, its implementation must occur within the ecosystem's capabilities (e.g., materials, resources, energy, land, and water) [13]. The economic aspect of sustainability emphasizes maximizing value and profit through proficiently using available resources without exploiting and damaging the ecosystem. On the other hand, the social aspect of sustainability concerns the public's welfare, such as in the interactions between an individual and the group (equity), public attentiveness and interconnection, and deployment of laborers and firms [13]. Accordingly, the Stockholm Resilience Centre categorized the 17 goals into three dimensions: (1) economic, which includes SDGs 8, 9, 10, and 12; (2) societal, which embraces SDGs 1, 2, 3, 4, 5, 7, 11, and 16; and (3) environmental, which involves SDGs 6, 13, 14, and 15 (Table 1) [18].

Although the literature has suggested that businesses are critical in achieving the SDGs, many have been reluctant to shift toward sustainable practices, technologies, and business models [16,19]. On the other hand, Lopez [20] posited that some businesses utilize a corporate social responsibility strategy to build a positive image and reputation. In contrast, other businesses respond to global demands by reflecting on the 17 SDGs through managing social and environmental projects. To evaluate the level of corporate sustainabil-

ity reporting practices within the SDGs, Tsalis et al. [21] suggested that it is very important to develop a framework. Previous studies in the systematic review predominately explored the role of business in the SDGs and focused on specific domains, such as B2B marketing, supply chains, and artificial intelligence [22–25]. To assist businesses and firms in incorporating the SDGs into strategic management practices, the United Nations Global Compact (UNGC) and the Global Reporting Initiative (GRI) have recently suggested that businesses "incorporate SDG reporting into their existing processes" [21]. Therefore, it is necessary to identify the overlooked areas of study among the 17 SDGs in the business and management discipline, as this will guide future studies. Thus, this study aims to: (1) explore the existing SDG studies in business and management and (2) identify the current stream of business and management studies that reflect SDGs. As this study is distinctive in providing an understanding of the current status of SDG progress, the findings of this study can provide future directions for SDG studies in business and management, which can fill a knowledge gap in these disciplines.

2. Materials and Methods

To achieve the objectives of this study, a Systematic Literature Network Analysis (SLNA) was conducted to understand the specific SDGs studied in business and management. SLNA, introduced by Colicchia and Strozzi (2012) [26], incorporates a systematic literature review and bibliographic network analysis, which is suitable for the subject under scrutiny, given its ability to grasp the limitless expansion of knowledge. Unlike a traditional literature review, SLNA offers the benefit of assessing literature using systematic and explicit methods to identify and critically evaluate relevant research topics, while citation network analysis detects emerging topics and analyzes the dynamics of knowledge evaluation [23]. By extracting quantitative information, SLNA helps depict the research areas that embody a delegation of the most relevant articles. The purpose of SLR is to find relevant research publications on a particular topic and to determine what can be reliably claimed based on these studies. SLR reveals the work that has already been accomplished in the subject area and the research approaches now being utilized [27]. Bibliometric networks demonstrate the relationships between publications, journals, researchers, and keywords. Common types of bibliometric network analysis are citation analysis, co-citation analysis, bibliographic coupling analysis, keyword co-occurrence analysis, and co-authorship networks analysis [28].

2.1. Systematic Literature Review

SLR was performed to define the scope and boundaries of the analysis so that it can trace studies according to "keywords, time, type of documents, language" [29,30]. The SLR procedure consists of three steps: (1) identifying the scope of analysis, (2) locating studies and filtering out "keywords, time, type of documents, and language," and (3) selecting and evaluating the relevant studies (Figure 1). Therefore, this study chose to perform the keyword search using the Web of Science (WoS) database because WoS is known as the world's dominant and highest-level database of scientific information [31]. Thus, the keyword search terms "sustainable development goals" and "sustainability" were used for this study in March 2022. Keywords were used, such as "TS = ("sustainable development goals" AND "sustainability"), and a total of 8,032 articles were collected. To locate relevant topics, we searched for articles in the business and management disciplines and articles with a social science citation index (SSCI) and a science citation index (SCI). Peer-reviewed articles were included, while other documents such as proceedings papers and early access articles were excluded. Only articles written in English were included. The year of publication was restricted to articles published from 2015 to 2021 since the SDGs were initiated in 2015. Through this process, 237 articles were collected and used for bibliometric network analysis.

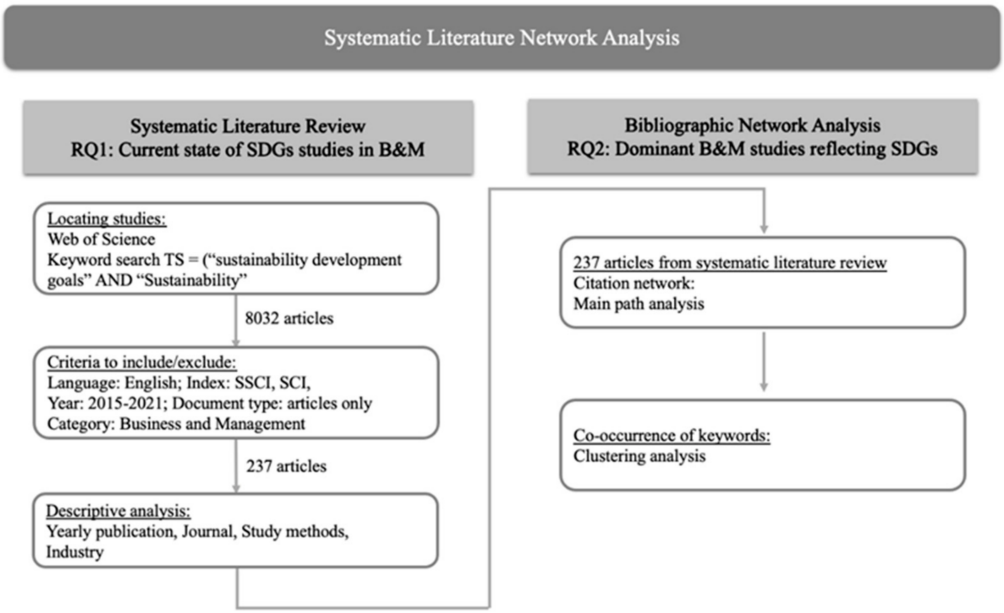

Figure 1. Outline of Systematic Literature Network Analysis.

2.2. Bibliometric Network Analysis

The BNA incorporates different types of analysis, such as citation network analysis, global citation scores, and keyword network analysis. Due to the objectives of this study, a citation network analysis (CNA) and a keyword network analysis (KNA) were conducted. The VOSviewer software was used for network visualization and the network analysis, while Pajek was employed to extract the main path. As an extension of the social network analysis, CNA helps visualize the research domains, how studies are interrelated in specific areas, and how knowledge structures are created in particular domains [26]. Since heavily cited articles function as hubs with many connections to other articles, citation networks can be analyzed according to clustering, reflecting, rising, and fading attention in research areas. As a bibliometric method, a main path analysis (MPA) can identify the primary streams of scientific knowledge and track the most significant paths for particular academic disciplines [32,33].

To detect patterns of research trends in Sustainable Development Goals, the keywords for the network analysis can be inputted, as this helps to recognize co-occurrences around the exact keywords or pairs of words that may denote a specific research theme. More specifically, the KNA can present the patterns and trends in a research field [31,34]. Therefore, co-occurrences in the analysis will detect the most frequent keywords, reflecting the conceptual knowledge structure of business and management's research on the SDGs.

3. Results

3.1. Descriptive Analysis

To comprehend the overview of the research on SDGs in business and management, SLR was employed, and a total of 237 articles were collected through a keyword search. The year of publication, country, journal distribution, and study methods were reported for a descriptive analysis.

3.1.1. Year of Publication

A total of 237 papers published between 2015 and 2021 were collected (Figure 2). Based on the yearly distribution of articles, the number of articles related to the SDGs in the business and management disciplines significantly increased after introducing the SDGs in 2015. A total of 10 articles were published in 2015, but the number of publications plummeted to 3 articles in 2016 before jumping back up to 26 in 2017. Since the United

Nations announced the 2030 Agenda for the SDGs, they have made serious efforts around the globe and even monitored Bangladesh after implementing the SDG national framework in 2016 [35]. In 2017, article publications related to the SDGs in business and management grew significantly, and the number of publications increased almost threefold from 26 to 75 articles in 2021. However, in 2018, the number of articles slightly decreased to 20 from 26 in 2017. Several factors, such an 8-10% drop in the stock market, the trade war between the U.S. and China, and the slowdown in global economic growth and rapidly increasing interest rates, caused an economic downturn in 2018 [36]. However, several experts and researchers are striving to develop a Statistical Data and Metadata Exchange (SDMX) standard for SDG data exchange, which Cambodia and Tanzania have used in pilot studies since 2019 [35]. More researchers from diverse disciplines, including business and management, also endeavor to develop specific indicators for each SDG so that they will all be feasible by 2030.

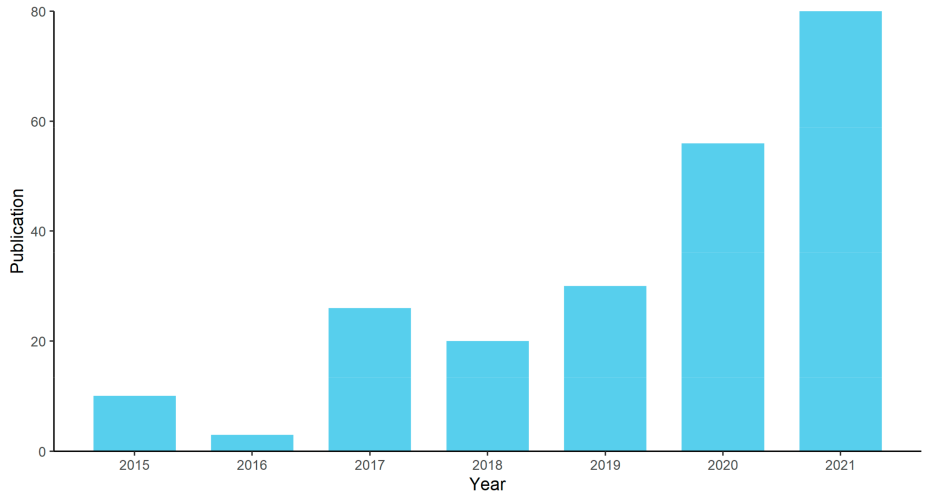

Figure 2. Distribution of the publications from 2015 to 2021.

3.1.2. Countries and Yearly Trends

Within three regions, 237 publications were developed in 59 countries (Figure 3). Research focusing on the SDGs in the disciplines of business and management predominate in three regions in Europe, North America, and Asia-Pacific, including 34 articles (14.35%) from the U.S., 31 (13.08%) from the U.K., 27 (11.39%) from Italy, 26 (10.97%) from China, 26 (10.97%) from Spain, 24 (10.13%) from Germany, 16 (6.75%) from Canada, 16 (6.75%) from France, 16 (6.75%) from India, and 14 articles (5.91%) from Australia (Figure 4). Although the U.S. published the most significant number of articles (34) related to the SDGs, approximately half (111, 46.84%) are from the European region, including the U.K., Italy, Spain, Germany, and France. This indicates that compared to North America, the European region is more proactive in addressing sustainability at various levels.

Most countries saw significant growth in the number of SDG articles, but the global pandemic caused a decline in SDG studies between 2019 and 2020 (Figure 5). The two leading countries, the U.S. and U.K., contributed the most sustainability-related publications, but these countries' research directions in SDG studies presented divergent trends. Since 2016, the U.S. has steadily increased its SDG research despite significant events such as Brexit, the exodus from the Paris Agreement, and the lockdown from the Covid-19 pandemic. On the other hand, the U.K. reported substantial declines in SDG research as they experienced economic and political issues stemming from Brexit and Covid-19. SDG research in other European countries, such as Italy and France, also significantly declined after 2017. In addition to the developed countries, China and India also started to produce SDG studies. Overall, the U.S., Spain, Germany, India, and Australia have steadily increased their output of SDG research and solutions in business and management since 2020.

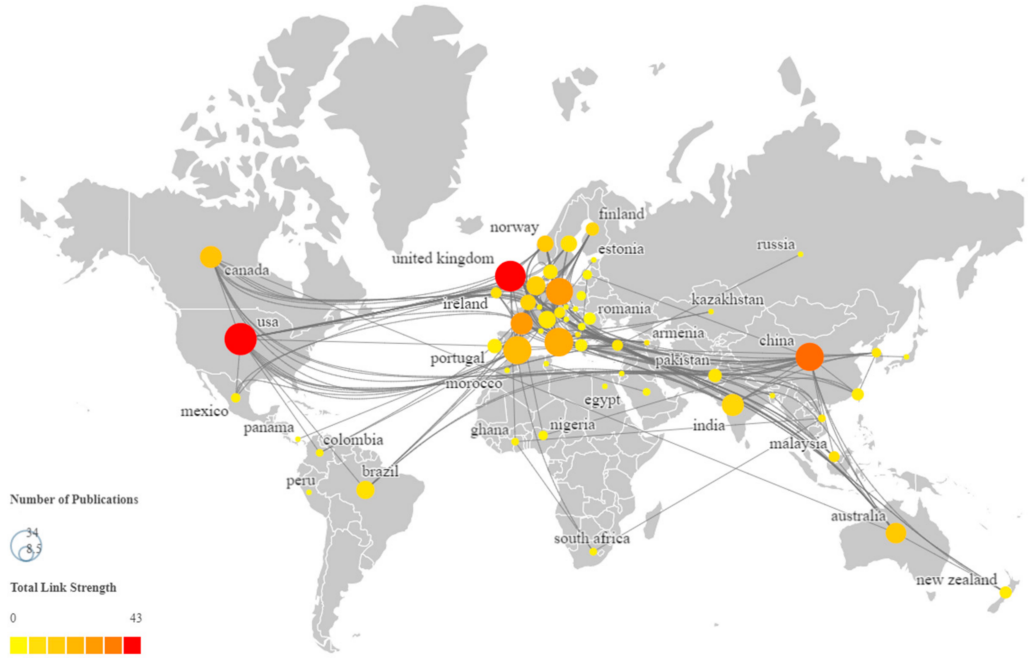

Figure 3. Distribution of publications within 59 countries/regions.

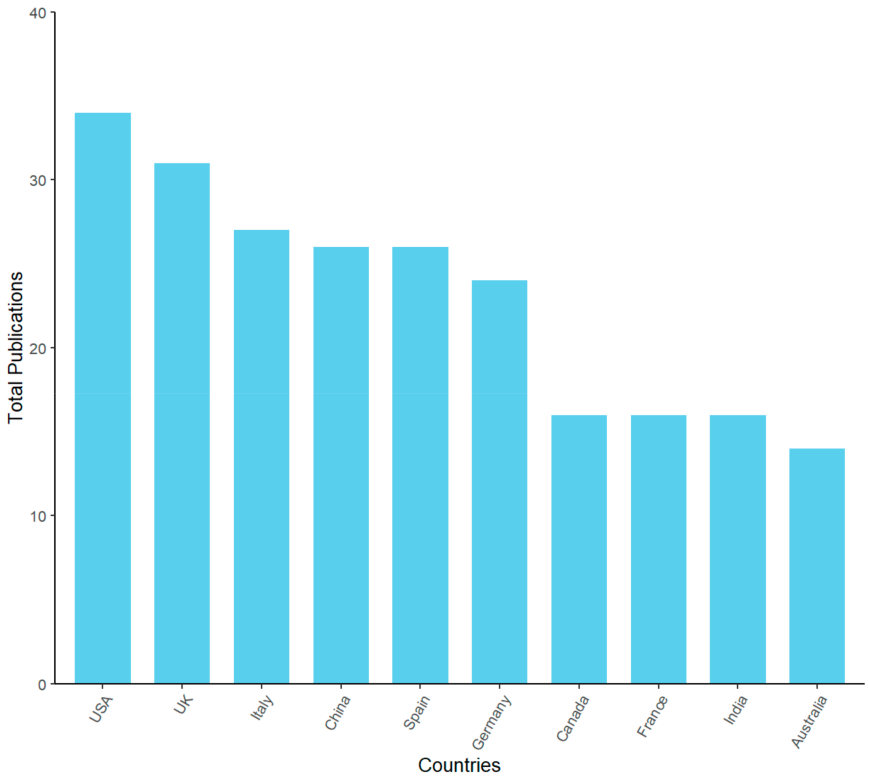

Figure 4. Publication of the ten most active countries/regions by year.

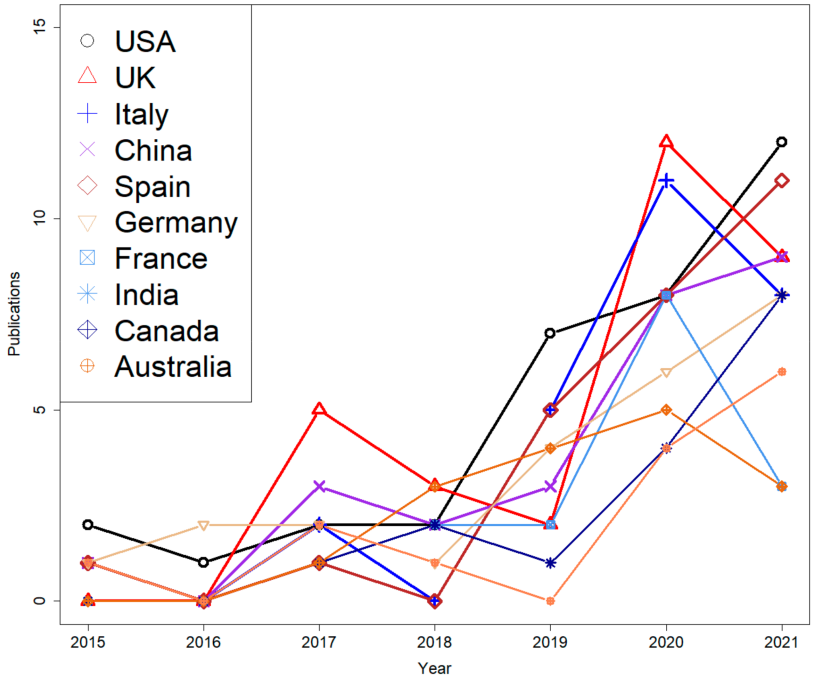

Figure 5. The most active ten countries/regions' publication trends by year (2015–2021).

3.1.3. Distribution of Papers by Journals

Among the 237 publications, a total of 74 journals were identified. About 53% of the articles were published in 10 journals: *Business Strategy and the Environment*, *International Journal of Management Education*, *Technological Forecasting and Social Change*, *Corporate Social Responsibility and Environmental Management*, *Sustainability Accounting Management and Policy Journal*, *Journal of Business Ethics*, *Marketing Intelligence Planning*, *Systems Research, and Behavior Science*, *Amfiteatru Economic*, and *Tourism Management Perspectives*. More specifically, a total of 27 (11.30%) articles were published in *BSE*, 23 (9.62%) from *IJME*, 21 (8.79%) from *TFSC*, 15 (6.28%) from *CSREM*, 12 (5.02%) from *SAMPJ*, and 7 (2.93%) from *JBE* (Figure 6).

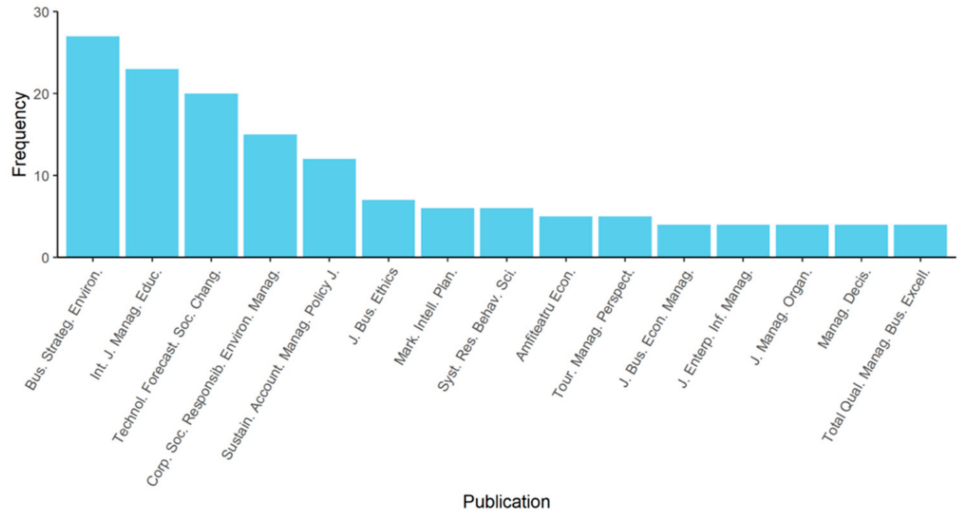

Figure 6. Distribution of journals.

3.1.4. Study Methods

Among the 237 articles, four qualitative, quantitative, conceptual, and mixed research approaches were identified (Figure 7). A total of 105 articles (44.3%) conducted qualitative research, followed by 90 quantitative articles (37.97%), 29 conceptual articles (12.24%), and

13 mixed methods research articles (5.49%). More specifically, among the 105 qualitative articles, different study methods were recognized, such as case study (39 articles), interview (22 articles), observation (4 articles), and focus group (4 articles) methods. The qualitative approach has been employed more frequently than the quantitative approach based on the ratio of study approaches. Among the quantitative articles, the most frequent study method used was a survey approach (48 articles). Quantitative studies are necessary to provide an effective framework and practical indicators. Without realistically measurable frameworks and indicators, it may be challenging to evaluate whether or not the SDGs have been met or need to be revisited. Through accurate measurement instruments, the UN's 17 goals and 169 objectives can be achieved by 2030 as planned.

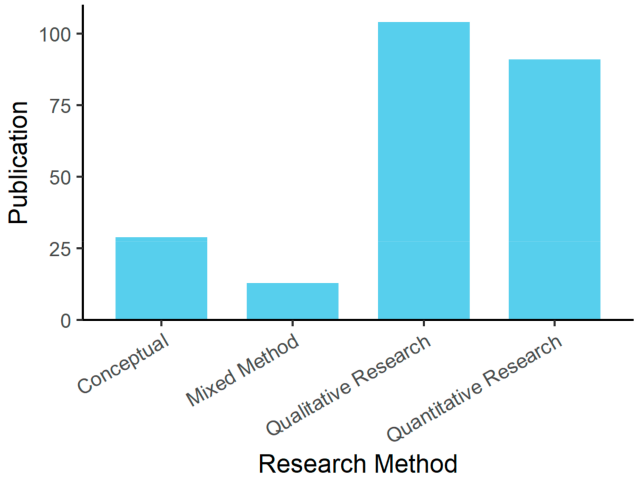

Figure 7. Distribution of research methods.

3.1.5. Industry Classification

Industries were classified into 24 areas (Table 2). Among the 237 articles, 93 (39.2%) were classified as general or/and non-specified, as these articles do not indicate a particular industry. Then, academia or education was reported, with 32 articles (13.5%) discussing the role of education and leadership in sustainable development. Followed by general and academia, 21 articles (8.9%) were related to the supply chain industry, and 20 articles (8.4%) were devoted to the energy industry. In addition, within the hospitality and tourism industry, the SDGs were the focus of 14 articles (5.9%).

Table 2. Distribution of industries.

Industries	Count of Papers	Percentage
General/Non-specified	93	39.2%
Academia/Education	32	13.5%
Supply Chain	21	8.9%
Energy	20	8.4%
Hospitality and Tourism	14	5.9%
Technology	9	3.8%
Food	8	3.4%
Manufacturing	8	3.4%
Healthcare	5	2.1%
Fashion	5	2.1%
Government	4	1.7%
Transportation	3	1.3%
Automobile	2	0.8%
Bank	2	0.8%
Construction	2	0.8%

Table 2. *Cont.*

Industries	Count of Papers	Percentage
Infrastructure	2	0.8%
Bioeconomy	2	0.8%
Agriculture	1	0.4%
Housing	1	0.4%
Military	1	0.4%
Public Health	1	0.4%
Recycling	1	0.4%

3.2. Bibliometric Network Analysis

3.2.1. Citation Network Analysis

The strength of a citation network analysis (CNA) is in identifying, analyzing, and visualizing the significant groups of articles that emerge from a specific topic or network (Van Eck & Waltman, 2013; Van Eck & Waltman, 2010). CNA counts the number of citations received by each publication to determine the most influential articles in a given field. A network is created by the nodes (e.g., articles) that are connected to the links (e.g., citations) [31]. More specifically, the results of CNA present the main flow of research and connections among studies, which can be easily observed through a map of the citation network of documents. To gather bibliographic data, VOSviewer software was utilized to visualize networks based on citations, which determines the relatedness of articles based on the number of times they cited each other [37]. Therefore, the VOSviewer was used to analyze and visualize the citation network of 237 publications (Figure 8). These studies examined a variety of issues, including the function and influence of sustainable education in the business and management fields [21,38–42], business strategies, and supply chains [43–46], and CSR and firm performance [19,47].

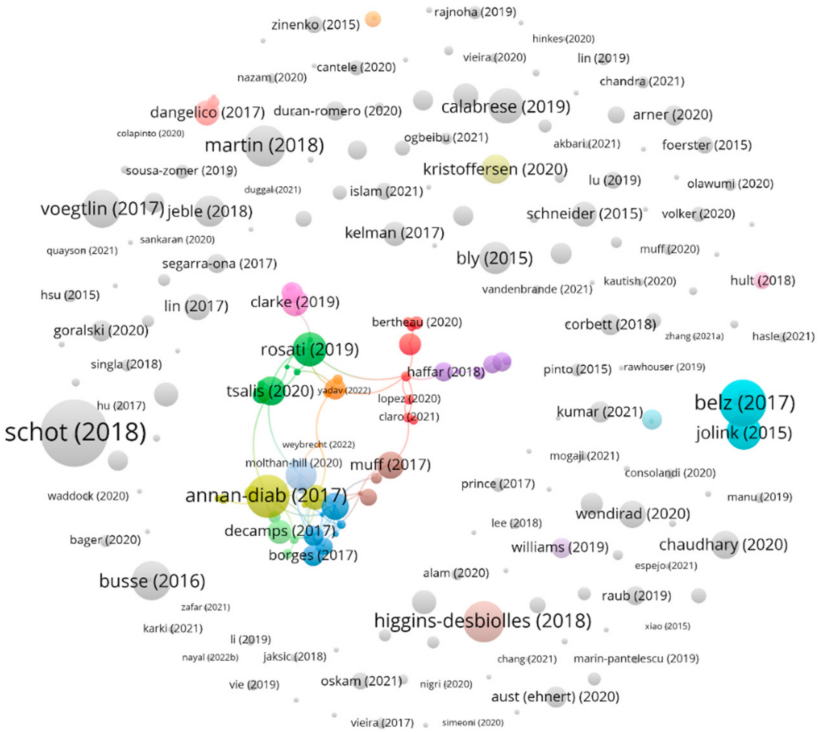

Figure 8. Citation Network Analysis [13,19,20,39,43,46,48,49].

3.2.2. Main Path Analysis

A main path analysis was conducted, as it helps identify the primary streams of scientific knowledge and the most critical paths for specific disciplines [32,33,50].

Hummon and Doreian (1989) [51] introduced a main path analysis for computing and analyzing the links (e.g., citations) between individual articles rather than the nodes (e.g., articles) in the citation networks. The main path analysis search method focuses on the connectivity of directed networks and finds a set of articles that have a significant impact on certain subjects [51]. In this regard, to complete the main path analysis, the software Pajek was used to analyze 237 articles in the area of business and management. Two significant elements of the Pajek software's main path analysis are traversal weights and the main path search because both calculate the relationships (links) between the articles (nodes) and discover the path with the highest value, respectively. Therefore, this study employed search path count (SPC) to calculate the traversal counts and a standard global search to determine the main path. The results revealed eight articles from 2017 to 2021 as the main path for all 237 articles (Figure 9).

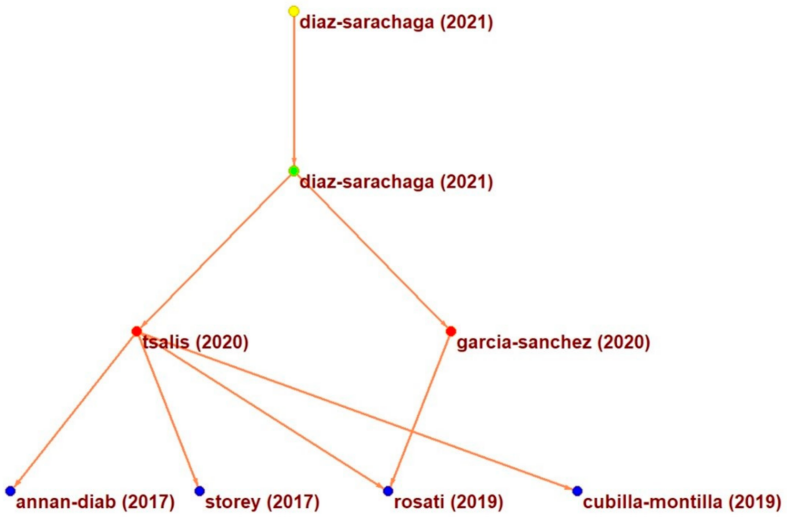

Figure 9. Main path analysis [19,21,38,41,47,52–54].

Storey, Killian, and O'Regan (2017) [41] reviewed the literature about Responsible Management Education (RME) and the Sustainable Development Goals (SDGs), the United Nations Principles for Responsible Management Education (PRME), and the sustainability context of RME in business schools. Initially, the study identified four different membership and affiliation organizations for business schools to engage with PRME: the Globally Responsible Leadership Initiative (GRLI), the Academy of Business in Society (ABIS), the Global Business School Network (GBSN), and the Environmental Association of Universities and Colleges (EAUC). Storey et al. (2017) [41] explained that a central role of RME is in leading practices, initiatives, and organizations in business schools, and the RME was dynamic and dispersed until 2017. In addition, the authors highlighted four teaching and learning initiatives, including "Giving Voice to Values," "Aim2Flourish organization," the future MBA, and the Sulitest measurement for PRME-related resources and practices. Similarly, Annan-Diab and Molinari (2017) [38] reviewed the literature on interdisciplinarity in sustainable development education with people from diverse backgrounds. The study demonstrated the significance of interdisciplinary approaches in sustainable development education, suggesting collaboration with professionals from various disciplines and sectors as a way to achieve the Sustainable Development Goals (SDGs).

On the other hand, Rosati and Faria (2019) [19] analyzed 408 worldwide organizations to determine the influence of adopting the SDG report. They grouped these organizations according to three criteria: (1) organization size, economic performance, and the presence of intangible assets (sustainable reputation or technological competencies) that can address the SDGs; (2) sustainability commitment and external engagement with voluntary sustainability programs and the use of external verification for sustainability management processes; and (3) corporate governance attributes to monitor the board of directors and shareholders. Accordingly, they found that early adoption of SDG reporting is associated

with large organizations, intangible assets, a solid commitment to sustainability frameworks and external assurance, a female board of directors, and a younger board of directors. Regarding corporate social responsibility reports, Cubilla-Montilla, Nieto-Librero, Galindo-Villardón, Vicente Galindo, and Garcia-Sanchez (2019) [54] found a vital role of the cultural values of a society on the social influence of business performance and firm sustainability reports. The cultural values they indicated included labor practices and decent working conditions, human rights, and social and product responsibility. Later, in investigating the effect of the disclosure of sustainable and corporate social responsibility (CSR) strategies on the investment decisions of investors, García-Sánchez, Rodríguez-Ariza, Aibar-Guzmán, and Aibar-Guzmán (2020) [47] found that foreign investors, pension funds, and other investors enhanced the relevance of sustainable and CSR information disclosures, while the government, financial institutions, and cross holdings were shown not to influence CSR disclosures.

Early studies [19,21,38,41,54] laid the groundwork for developing an evaluation framework for sustainable development assessment that can measure the quality of the information in corporate sustainability reporting based on the SDGs. These researchers found that firms tend to release information regarding sustainable energy strategies, sustainable investments, and climate change-related actions and strategies, but are unwilling to disclose the erasure of corruption incidents and human rights violations. Moreover, they found that the industry sector's characteristics can affect the corporate sustainability report. Specifically, industry sectors such as financial services, retailers, aviation, energy utilities, metal products, energy, mining, construction materials, and telecommunications were not inclined to provide comprehensive information in their sustainability reports.

Based on a previous study by Tsalis et al. (2020) [21] and García-Sánchez et al. (2020) [47], Diaz-Sarachaga (2021) [53] identified inconsistencies between the global reporting initiative (GRI) declarations and the sustainability performance of Spanish companies. They addressed that the primary characteristics of corporate reporting by Spanish companies featured intangibility, omission of negative repercussions, inadequate standardization, variability of criteria, and lack of comparability. After these results, Diaz-Sarachaga (2021) [52] emphasized the feeble correlation between several SDG indicators and company activity, as well as the difficulty of disaggregating accounting information. They also pointed out that unclear and unaccountable SDG indicator descriptions still existed. Instead, some indices, such as climate change management, poverty reduction, economic growth, and environmentally sound technologies, were highly concentrated.

3.2.3. Keyword Analysis

To identify research trends on the SDGs, a keyword analysis was conducted, as keyword co-occurrence can be classified by the scrupulous associations that emerge [55]. Distinct from citation analysis, the analysis of the co-occurrence of keywords does not contain an intrinsic bias toward older studies but instead encourages important recent works to emerge [7]. As a reference point in finding and understanding the central content of articles, keywords locate the progress of the research field over time [56]. Therefore, due to the SDGs being relatively new but subject to rapid technological transformation, it is essential to determine the co-occurrence of keywords that complement the citation analysis by recognizing the main topics and trends through clustering. Using 237 articles, keyword analysis was conducted to understand how a network of co-occurring keywords creates clusters for SDG studies.

Before proceeding with the keyword analysis, keywords were screened out for duplicates, normalized in spelling, and unique values were used for the analysis [57]. Keywords that were encountered at least five times were retained. Data were analyzed with VOSviewer to investigate the co-occurrence of keywords, resulting in 72 items and 916 links (Figure 6). The co-occurrence of keyword analysis detected the high-frequency keywords in the dataset which were "sustainability" (frequency: 86), "sustainable development" (frequency: 69), "corporate social responsibility" (frequency: 55), "sustainable development goals" (frequency: 52),

"management" (frequency: 39), "performance" (frequency: 38), "innovation" (frequency: 23), and "corporate sustainability" (frequency: 23). Other high-frequency keywords include "business" (frequency: 21), "impact" (frequency: 21), "governance" (frequency: 19), "green" (frequency: 14), "climate change" (frequency: 14), "system" (frequency: 14), "technology" (frequency: 13), "design" (frequency: 10), "environment" (frequency: 6), and "energy" (frequency: 6).

Based on the networks of keywords, clustering presents articles that are grouped, which allows for the allocation of topic areas. The VOSviewer clustering technique detected the 72 research keywords and categorized them into five clusters. Similar studies were grouped to determine the relatedness between various keywords based on the frequency with which they occurred together. Five clusters were classified as technology and innovation, education and human resource management, CSR and firm performance, governance, and business strategies and supply chains (Figure 10). The more significant the cluster, the greater the number of keywords, and the relatedness of keywords is displayed by different visualizations, such as the distance between keywords and the curved lines connecting them. Regarding citations, the distance between two keywords represents relationships in which they are near other keywords, and their citations are highly connected. The curving lines between the keywords depict the relationship between them, with the thickness of each line denoting the number of sources between each pair.

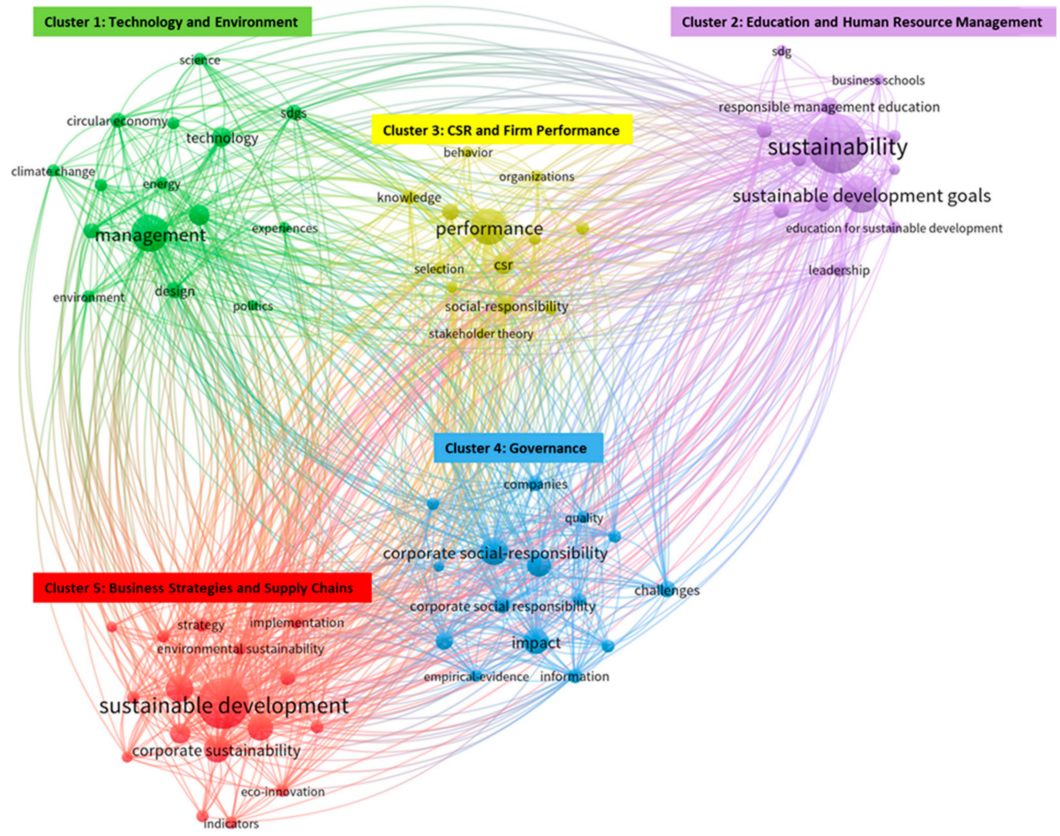

Figure 10. Clusters of the co-occurrence network of keywords.

Technology and environment, Cluster 1 (Green), comprises articles related to technology and environmental perspectives on the SDGs. The most frequently used keywords include "innovation," "technology," "energy," "environment," "climate change," "design," "circular economy," "management," and "systems." The literature in this cluster discussed sustainable innovation policies, strategies, and technology and technology policies across different levels. Förster (2015) [58] suggested that automobile manufacturers should consider implementing environmentally sustainable technologies or process cost reductions through resource or energy conservation and customer and legislative demands for a

high level of sustainability. Förster also emphasized that large automakers are expected to adopt new sustainable technologies sooner than small- and medium-sized businesses. Focusing on the SDGs' innovation, Voegtlin and Scherer (2017) [49] described three aspects of responsible innovations for sustainable development, highlighting people and plants, doing good, and practicing accountable governance.

Aligning with Voegtlin and Scherer's (2017) [49] view, Schot and Steinmueller (2018) [48] underlined the importance of transformative and sustainable innovation policies and the current high demand for transformative change in a variety of socio-technical systems. To initiate sustainable innovations, de Medeiros, Vidor, and Ribeiro (2018) [59] stressed leaders' proactive roles and that investments in environmental innovations are essential for industrial organizations. Supporting Förster's (2015) [58] study, Doluca, Wagner, and Block (2018) [60] indicated that small enterprises and family firms are relatively delayed in adopting sustainable innovations (e.g., the green design of products and cleaner technology), but that these firms are prone to being more stable as compared to other types of firms. On the other hand, Managi, Lindner, and Stevens (2021) [61] addressed various science, technology, and innovation (STI) policies that affect the decision-making process for implementing the SDGs at different levels (local and global). More importantly, they emphasized that the effect of the SDGs on innovation is context-dependent and that the relationship between STI and the SDGs is strongly associated at different levels (local, regional, and national). This cluster also discusses individual levels of acceptance of technologies for sustainability. According to Hua and Wang [62], ease of use has a significant impact on consumers' perceived usefulness, which in turn has an impact on customers' attitudes and intentions to utilize energy-efficient technologies. Asadi et al. (2019) [63] integrated the norm activation theory and the theory of planned behavior to identify several key factors (awareness of consequences, attribution of responsibility, subjective norms, personal norms, attitudes toward green IT, cost savings, competitive advantage) influencing decision makers' environmental intentions toward Green IT adoption in the manufacturing industry. In addition, climate change beliefs have a substantial positive relationship to personal environmental norms and environmentally conscious behavior, and several authors suggested enhancing personal environmental norms to increase such behavior [63]. Moreover, one study explored the influence of personal values and political structure on the sustainable air travel behavior of individuals [62]. The results revealed that internalized knowledge about climate change and the impact of air travel is essential for customers' reduction in air travel behavior [62].

Education and human resource management, Cluster 2 (Purple), contains articles focusing on the importance of education in business and management. The most frequently mentioned keywords in this cluster include "sustainability," "sustainable development goals," "leadership," "education," and "responsible management education." This cluster addressed the role of business schools in delivering sustainability-related management education and encouraging future business and management leaders to make responsible and sustainable decisions. Considering the critical role of management education in achieving the SDGs, Storey et al. (2017) [41] reviewed the literature on responsible management education (RME) in SDG-related domains, highlighting that the contemporary RME field is dynamic but presents a trend of consolidating around the theme of the SDGs (UN PRME). On the other hand, Annan-Diab and Molinari (2017) [38] proposed interdisciplinary education methods for sustainable development, specifying that these should consist of multiple disciplines collaborating in an integrated manner. They emphasized that a multidisciplinary approach to education for sustainable development is practical and helpful in addressing complicated sustainable development challenges for both instructors and students. However, Weybrecht (2017) [64] pointed out the many challenges of management education in the context of the SDGs because sustainable education in business schools is somewhat overlooked in the current sustainable development literature.

CSR and firm performance, Cluster 3 (Yellow), focuses on corporate social responsibility (CSR) and firm performance related to sustainability challenges. "Performance,"

"corporate sustainability," "CSR," "organization," "stakeholder theory," and "business" were the most frequently used terms. Ahmad and Mehmood (2015) [9] investigated the association between sustainable enterprise and supply chain management systems in smart cities. They found that the motivational factors can be categorized into environmental, business, and technical aspects for firms and cities and can contribute to the firm's performance. To establish a sustainable performance evaluation for companies, Hsu, Ou, and Ou (2015) [65] proposed sustainable business performance evaluation criteria by establishing financial metrics (e.g., operating ability, solvency, and profitability), credit risk, and environmental and social responsibility.

On the other hand, Schneider (2015) [66] suggested that it is essential to consider reflexivity in the relationship between corporate sustainability goals and the overarching objective of sustainable development. Moreover, by conducting grey relational analysis, they proposed a modified TOPSIS method developed as a ranking methodology for many real-world applications in science and engineering for sustainability performance evaluation. Doluca, Wagner, & Block (2018) [60] investigated the impact of environmental practices and initiatives in four-wave surveys and found that family firms were reluctant to initiate environmental activities and innovations early. They also learned that family firms' environmental behavior is less fickle and more committed over time than non-family firms. On the other hand, Rosati and Faria (2019) [19] suggested that larger organizations and greater levels of intangibility were associated with better firm performance.

Cunha et al. (2020) [67] addressed the presentation of heterogeneous phenomena in sustainable investment performance but also emphasized an optimistic prospect for superior risk-adjusted returns in certain regions by analyzing the performance of the Dow Jones Sustainability Indices. Supporting Cunha et al.'s (2020) [67] results, Khan, Johl, and Johl (2021) [68] suggested that adopting a sustainability index and green innovation practices could enhance firms' SDG performance and inspire greater stakeholder confidence while increasing the level of transparency in business activities. More specifically, Tsalis et al. (2020) [21] proposed a methodological framework to assess current corporate sustainability practices within the scope of the SDGs. A sample of 48 sustainability reports found that firms provide more information about their actions on issues related to their strategies for energy reduction, infrastructure and innovation, and climate change based on the 17 SDGs. In addition, García-Sánchez et al. (2020) [47] pointed out that ownership by foreign investors, pension funds, and other investors was shown on sustainability reports from 989 international companies, while government, financial institutions, and cross holdings did not have much impact on the information systems related to the 2030 SDG Agenda.

Governance, Cluster 4 (Blue), concentrates on articles that discuss the government's involvement, governance, and policies toward the SDGs. The most frequently used words in this cluster include "governance," "companies," and "impact." Environmental, social, and governance (ESG) is a term that frequently appears in this cluster, referring to corporate decisions or investments that take environmental, social, and governance concerns into account [69]. Considering that the ESC plays a vital role in investment decisions, Friede (2019) [69] analyzed ESG integration barriers by reviewing 112 previous studies that featured 10,600 interviews and survey responses. In turn, by conducting a textual analysis, Friede (2019) [69] identified 161 topics classified into market, firm, regulatory, and individual-based impediments. By using health care companies as a case study for SDG 3, Good Health and Eell-being, Consolandi, Phadke, Hawley, and Eccles (2020) [70] argued that SDGs have predominantly focused on the social and environmental aspects of sustainability. They also pointed out that the SDGs target the Sustainability Accounting Standard Board's (SASB's) 30 generic ESG issues in considering both financially material and irrelevant ESG issues based on SASB.

Regarding the role of government in achieving the SDGs, Voegtlin and Scherer (2017) [49] investigated how responsible innovation contributes to sustainable development. They implicated global governance plans based on discussion to substitute responsible innovations (e.g., business organizations resolving sustainability challenges, regulating

responsible innovations without harm, and creating sustainable societies for business innovations). Lin (2019) [71] examined whether or not the role of government–business partnerships (GBP) is practical and valuable for programs targeting reflective environmental impacts via the promotion of radical eco-innovation. Based on 166 U.S. firms in 192 environmental alliances between 1985 and 2013, the author addressed four fundamentals of GBP actions, such as effective governance, exploration learning, cognitive learning, and rulemaking, that embrace radical environmental solutions. Moreover, Lin (2019) [71] suggested that GBP guides public and private efforts in tracking transformative environmental change by adopting radical eco-innovations.

Business strategies and supply chains, Cluster 5 (Red), includes articles deliberating business plans aligned with the SDGs and sustainable supply chain practices. "Sustainable development," "strategy," "environmental sustainability," and "sustainable entrepreneurship" are the most frequently used terms in this cluster. To discover the contextual barriers to supplier development of sustainability in global supply chains, Busse, Schleper, Niu, and Wagner (2016) [43] conducted interviews with Western European buyers and Chinese suppliers, supplemented with 81 documents. They explained the four types of contextual barriers to sustainability as complexities in the sustainability concept, socio-economic differences, spatial and linguistic distance, and cultural differences between buyers and suppliers. To bridge these divides, effective joint communications, an open organizational culture, and the fostering of cross-contextual understanding were suggested [43]. Moreover, Jeble et al. (2018) [44] examined the influence of big data and predictive analytics on the sustainable business development goals of the organization in terms of environmental, social, and economic performance. Considered one of the most critical capabilities, big data and predictive analytics consist of tangible resources for technical skills, management skills, organizational learning, and data-driven culture, and play an imperative role in companies' sustainable competitiveness and achievement of the SDGs.

4. Discussion

Sustainability is a growing social concern, and numerous businesses have voluntarily implemented sustainability practices since early 2000, while the United Nations has stressed its importance by proposing the Sustainable Development Goals (SDGs). Specifically, the UN's SDGs were established in 2015, focusing on 17 goals with 169 objectives that aim to be instituted by 2030. Since then, many researchers have explored how the SDGs can be measured and evaluated so that they can suggest methods for implementation in businesses, organizations, and management. Therefore, this study purports to explore the existing concentrations of SDG and sustainability-related studies in business and management that can foresee the future directions of SDG research by understanding the current trends. The objectives of his study were to comprehend SDG research in the business and management disciplines and to recognize the current areas of business and management studies reflecting SDGs. A systematic network literature review of 237 articles was conducted and analyzed using a systematic literature review (SLR) and bibliographic network analysis (BNA) through a keyword search to provide a future direction for SDG studies.

The findings from the SLR indicate the following overview of trends in SDG studies in business and management: (1) they have continuously increased since 2015; (2) the majority of studies were published in Europe and North America; and (3) the predominant research approaches were qualitative. Based on yearly trends, it was shown that there was a slight dip in 2018 but steady growth since 2019. Due to the global economic downturn in 2018, article publications related to SDG studies were impacted. However, after 2018, research related to the SDGs increased to 75 articles by 2021. Moreover, most articles originated from Europe and North America and were from countries considered developed (e.g., the U.S., U.K., and Italy). Similar to yearly publications, 2018 saw a slight drop in SDG research but then it quickly increased again. Most notably, China and India produced substantial research related to the SDGs during this time. Since these two countries rely highly on manufacturing sectors as a part of the supply chain, buyers from developed

countries might require manufacturing sectors to engage in sustainability. On the other hand, a predominant number of research approaches were reported as being qualitative over quantitative. This implies that most research is still exploring the SDGs to provide a better framework and indicators to measure, assess, and evaluate them. According to the findings from the SLR, research related to the SDGs is continuously increasing but still needs to explore measurable and evaluative ways to achieve the SDGs by 2030.

The findings from the BNA presented five clusters categorized as technology and environment, education and human resource management, CSR and firm performance, supply chains and governance, and business strategies. Keywords in the technology and environment cluster include previous literature reflecting SDG 9, Industry, Innovation, and Infrastructure. Classified as the economic aspect of sustainability by the Stockholm Resilience Centers (Table 1), the predominant literature [48,49,58,60,61] addressed how different types of innovations, such as the design of products and environmentally sustainable technologies and processes, could impact the level of sustainability and decision-making processes on both the local and global levels. The education and human resource management cluster assembled literature on leadership, education, and sustainable management education, which suggests SDG 4, Quality Education. Considering that SDG 4 reflects the social aspect of sustainability (Table 1), the literature in this cluster discussed responsible management education in the different disciplines, interdisciplinary methods, and different challenges of management education in the context of the SDGs [38,41,64]. The CSR and firm performance cluster present articles relating to performance, corporate sustainability, and CSR, replicating SDG 8, Decent Work and Economic Growth. As SDG 8 concentrates on the economic aspects of sustainability (Table 1), previous studies [9,19,60,65–67] have cataloged the influence of CSR, corporate sustainability, and the sustainability index on firm and SDG performance.

Moreover, the governance clusters feature articles concerning governance, policies, and government-business partnerships, which reflect SDG 16, Peace, Justice, and Strong Institutions. Based on criteria from the Stockholm Resilience Centers (Table 1), SDG 16 fulfills the social aspect of sustainability. Most studies focus on how the environmental, social, and governance factors, such as global governance, result in sustainable development to create sustainable societies and eco-innovations [49,69–71]. Lastly, the business strategies and supply chain cluster contain articles discussing sustainable entrepreneurship, design, sustainable development, and environmental sustainability. Therefore, representing SDG 12, Responsible Consumption and Production on the economic aspect of sustainability (Table 1), the previous literature has discussed supply chains' contextual barriers to sustainability and information technology and the impact of big data on companies' sustainable competitiveness and implementation of the SDGs [43,44].

In summary, the findings from the BNA revealed five clusters that target three aspects of sustainability, economic, social, and environmental, aligning with the five different SDGs. This implies that the business and management disciplines must adopt and consider all aspects of sustainability to achieve the UN's 17 SDGs. Moreover, this study discussed five SDGs (e.g., SDG 4, 8, 9, 12, and 16) aligned with five clusters which connect to future research concentrating on the remaining SDGs (e.g., SDGs 3, 5, 6, 7, 10, 11, and 13). Due to the growing public interest in sustainability, as well as the UN's 2030 goal, research on the remaining SDGs should be accelerated. Specifying accountable, measurable, and evaluative frameworks and indices for the SDGs is necessary.

5. Conclusions and Limitations

This study aims to understand the progress of Sustainable Development Goal studies in business and management by exploring the existing SDG studies from 2015 to 2021. By conducting a systematic literature network analysis, this study helps to understand the concentrated areas of SDG studies, such as technology and environment, education and human resource management, CSR and firm performance, governance and business strategies, and supply chains. Likewise, the findings of this study provide valuable insights

to illustrate an overview of the extant SDG studies, but there are some limitations. This study only explored SDG studies in the business and management disciplines, which omits other aspects of SDG studies. To achieve the SDGs by 2030, many different industries and disciplines have put tremendous efforts into exploring the SDGs. Therefore, future research can be expanded to more fields so that the 17 SDGs can be explored in biology, agriculture, agrobiology, and engineering. Moreover, this study employed a systematic literature network analysis, which collects and synthesizes articles that fit prespecified eligibility criteria. Future research can pursue a meta-analysis focusing on statistical methods to summarize the results of these studies. As this study only included articles from 2015 to 2022 due to the SDGs being established in 2015 by the UN, there is a limitation to understanding the overview of sustainability studies published earlier than 2015. Therefore, future research can expand into earlier years exploring sustainability to understand the overarching trends before the SDGs were established. Moreover, this study did not distinguish between business and management disciplines' public and private sectors. Therefore, future research can explore how SDG studies have investigated the differences between these sectors.

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